

- 5 There is a global shortage of blood for transfusions. Researchers can culture bone marrow stem cells in the laboratory to manufacture red blood cells for potential use as an artificial blood product.

The researchers collect bone marrow stem cells that are present in small quantities in blood, rather than extracting them from bone marrow.

- Antibodies, specific to bone marrow stem cells, are attached to tiny magnetic beads.
 - The beads are added to a sample of blood.
 - An electric field is applied to immobilise the beads so that the bone marrow stem cells can be collected.
- (a) The antibodies are attached to the beads so that the sites used for binding to specific molecules on the bone marrow stem cells are left exposed.

Name the term given to the specific molecules on the bone marrow stem cells that attach to the antibody binding sites.

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- (b) Suggest **and** explain the advantages of using bone marrow stem cells from the blood sample to manufacture artificial red blood cells.

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- (c) One desirable feature of artificial blood products, such as artificial red blood cells, is that they should be economical to produce.

Suggest **other** desirable features of artificial blood products.

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[Total: 7]